

WEEK 1

PROGRAM 1

```
#include<iostream>

using namespace std;

int main(){

    cout<<"Zahurunnisa Begum"<<endl;

    cout<<"CSE "<<endl;

    cout<<"25A31A05EI"<<endl;

    cout<<"SEC-D"<<endl;

    return 0;

}
```

PROGRAM 2

```
#include <iostream>

using namespace std;

int main() {

    double num1, num2, num3;

    cout << "Enter three numbers: ";

    :cin >> num1 >> num2 >> num3;

    if (num1 >= num2 && num1 >= num3) {

        cout << "The biggest number is: " << num1 << std::endl;
```

```

    }

    else if (num2 >= num1 && num2 >= num3) {

        cout << "The biggest number is: " << num2 << std::endl;

    }

    else {

        cout << "The biggest number is: " << num3 << std::endl;

    }

    return 0;

}

```

PROGRAM 3

```

#include <iostream>

using namespace std;

int main() {

    char op;

    double num1, num2;

    cout << "Enter an operator (+, -, *, /): ";

    cin >> op;

    cout << "Enter two operands: ";

    cin >> num1 >> num2;

    switch (op) {

```

case '+':

cout << num1 << " + " << num2 << " = " << num1 + num2;

break;

case '-':

cout << num1 << " - " << num2 << " = " << num1 - num2;

break;

case '*':

cout << num1 << " * " << num2 << " = " << num1 * num2;

break;

case '/':

if (num2 != 0) {

cout << num1 << " / " << num2 << " = " << num1 / num2;

} else {

cout << "Error! Division by zero is not allowed.";

}

break;

default:

cout << "Error! The operator is not correct.";

break;

}

cout << endl;

```
    return 0;
}
```

PROGRAM 4

```
#include <iostream>

using namespace std;

int main()
{
    int n;

    cout << "Enter a number: ";

    cin >> n;

    cout << "Prime numbers up to " << n << " are: ";

    for (int i = 2; i <= n; i++)
    {
        bool isPrime = true;

        for (int j = 2; j <= i / 2; j++)
        {
            if (i % j == 0)
            {
                isPrime = false;

                break;
            }
        }
    }
}
```

```

    }

    if (isPrime)
    {
        cout << i << " ";
    }
}

return 0;
}

```

PROGRAM 5

```

#include <iostream>

using namespace std;

int main() {

    int n;

    long long factorial = 1;

    std::cout << "Enter a positive integer: ";

    std::cin >> n;

    if (n < 0) {

        cout << "Factorial of a negative number doesn't exist." <<
std::endl;

    } else {

        for (int i = 1; i <= n; ++i) {

            factorial *= i;

```

```

    }

    cout << "Factorial of " << n << " = " << factorial << std::endl;

}

return 0;

}

```

PROGRAM 6

```

#include <iostream>

using namespace std;

long long factorial(int n) {

    if (n == 0 || n == 1) {

        return 1;

    }

    else {

        return n * factorial(n - 1);

    }

}

int main() {

```

```
int number;

cout << "Enter a non-negative integer: ";

cin >> number;

if (number < 0) {

    cout << "Factorial is not defined for negative numbers." <<
std::endl;

} else {

    long long result = factorial(number);

    cout << "Factorial of " << number << " is: " << result << std::endl;

}

return 0;

}
```